GHS cover census at larger q (n = 3) and for n = 5

2026-10-14. Conductor T-2467. Extends research/ghs_cubic_cover_census_20261012/ (PR #1759, "the merged census"). **Structural stage diagnostic on toy fields.** It counts covers by Riemann–Hurwitz genus. No curve over F_q is constructed, no transfer map is evaluated, no discrete logarithm is solved, and no speedup is claimed.

Protocol: PROTOCOL.md, committed (2d95bde3f) before any panel ran, with one amendment (338cd011a, hypothesis H) added after panel D and before panels A, B, C, E produced output; protocol-versioned at commit, not independently preregistered. Run 20261014T0050Z: records, environment and timings, reproduction checks, run report. This report as PDF: REPORT.pdf. Tool: cover_census_n.rs (dependency-free Rust). Lean: lean/CoverCensusN5.lean.

Question

For E: $y^2 = x^3 + a_2x^2 + a_4x + a_6$ over K = F_{q^n} and the x-line with branch set B = x(E[2]), the smallest genus of the (Z/2)^m compositum of the n conjugate double covers over every F_q-structure τ (g = 2^{m−2}(r−4)+1). (1) Does the merged n = 3 pattern — class minimum 13 / 5 / 3 for #E(K) odd / 2 mod 4 / 0 mod 4 — hold at larger q? (2) For n = 5, what is the minimal genus and what governs it?

Result in brief

- **The new enumeration reproduces the merged census exactly**: every q = 5, 7 curve record and mod4 histogram, and the q = 11 sample. At n = 5 it agrees with the generalised cocycle search on every curve.
- **n = 3:** at q = 11 (2 662 curves, exhaustive) the class pattern 13 / 5 / 3 holds on every curve. At q = 13 (4 400 curves, exhaustive) **the pre-registered prediction fails in one row**: 2 of 1 100 curves of order 2 mod 4 have class minimum 9. Both lie in the trace-zero class #E(K) = q^3 + 1, as do the merged census's two q = 5 exceptions (post-hoc observation). The q = 17 sample (200) is consistent. New theorem: **#E(K) ≡ 2 mod 4 ⟹ genus ≥ 5 on every structure, for every q.** So "2 mod 4 never reaches 3", previously only observed, is now proved; the odd-order bound of 13 was already proved.
- **n = 5:** the minimal genus is governed by the 2-torsion type, with derived floors **33 (odd) / 33 (2 mod 4) / 25 (E(K)[2] = Z/2, 4 | #E) / 5 (E(K)[2] = (Z/2)²)**. Further theorems: m ∈ {1, 4, 5}; no genus in 6–16; for odd order the minimum is 33 exactly when j ∉ F_q. All floors are attained at q = 3 (exhaustive, 488 curves). Genus 5 occurs on 20 of 81 split curves at q = 3 and on none of the 31 split models censused at q = 5 (sample). 0 violations of any derived statement, in any arm.

How the census is computed (new enumeration)

Every structure is τ = MσM^{−1} (Lang), one per coset M·PGL2(F_q), and τ^iB = Mσ^i(M^{−1}B), so the genus of (B, τ) is that of (B' = M^{−1}B, σ). The tool runs N = M^{−1} over a transversal of PGL2(F_q)\PGL2(K) given by normal forms of (N(∞), N(0), N(1)) and takes σ^iB' as coefficient-wise Frobenius images; a gcd test (gcd(B', σ^kB') = 1, k ≤ (n−1)/2) settles the pairwise-disjoint structures (m = n, r = 4n) without forming unions. Cost per curve $(Q^3 - Q)/(q^3 - q)$ instead of the merged search's Q^3 per field and |PGL2(K)|-scan per curve. The merged cocycle search, generalised to A·A^σ⋯A^{σ^{n−1}} scalar, is kept as a second method.

Reproduction (all exact, checks):

check	what was compared	result
R1	n = 3, q = 5, 7, every curve: A_idx, B_idx, order, maxprime, min_genus, types_at_min, structures_by_genus, m1_structures, min_genus_2isog, min_genus_class against merged run 20261013T0403Z; and its mod4 histograms	identical, 252 + 690 curve records, 9 + 9 histograms
R2	n = 3, q = 11, the merged 120-curve sample (same seed and stream), same fields minus class minimum	identical, 120 records
R3	n = 5, q = 3, every curve: coset method against the generalised cocycle search (all fields but mismatch)	identical, 488 records
pilot	n = 3, q = 3, both methods	identical, 56 records

The subcover-sum identity (genus = sum of (s−2)/2 over the nonzero vectors of the span) held on every structure evaluated in full in every arm (mismatches = 0), and every curve counted exactly $(Q^3 - Q)/(q^3 - q)$ structures (asserted by the tool).

Panels

panel	n	q	coverage	models	structures per model	wall time
R1	3	5, 7	every class	252, 690	16 275, 120 099	6 s, 42 s
R2	3	11	merged 120-curve sample	120 + 127 neighbours	1 786 323	142 s
A	3	11	every class	2 662	1 786 323	918 s
B	3	13	every class	4 400	4 855 539	4 956 s
C	3	17	sample of 200	200 + 187 neighbours	24 221 379	1 245 s
D	5	3	every class	488	597 861	98 s (cocycle method, R3: 3 232 s)

panel	n	q	coverage	models	structures per model	wall time
E	5	5	sample of 60	60 + 64 neighbours	254 313 125	4 746 s (after one restart, below)
pilot	3	3	every class (char 3)	56	819	< 1 s

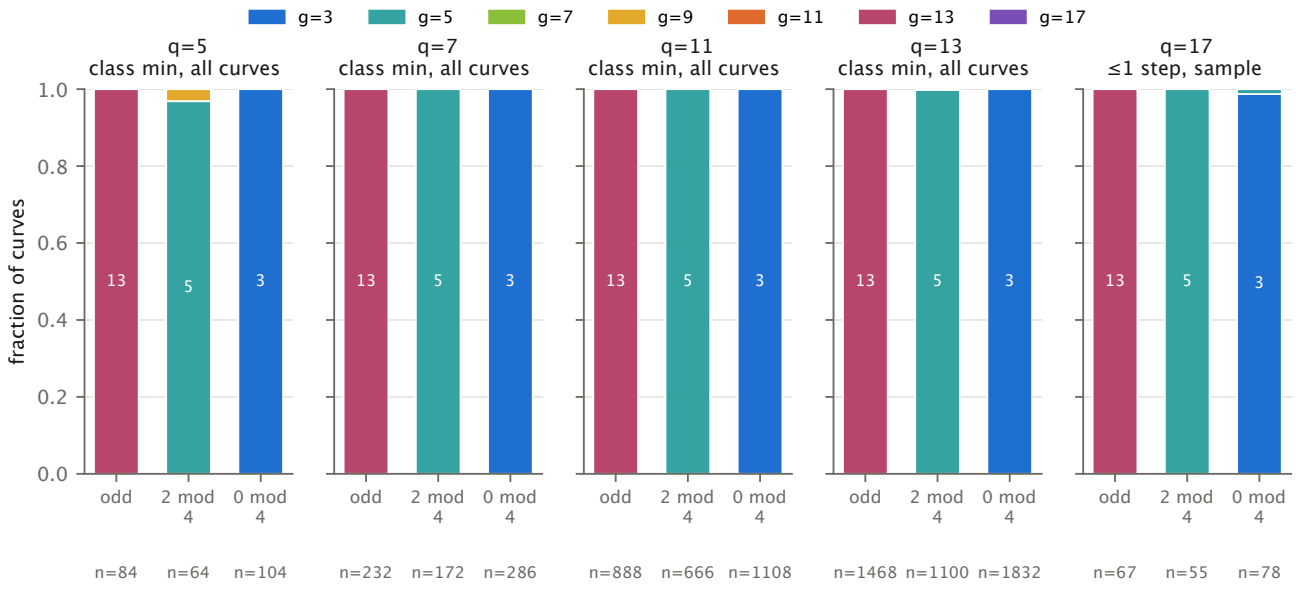
Wall times on a shared host at load average 60–260 (6 threads), bookkeeping only. Panel E was killed once when the host restarted, about 2 h 15 min in, before it had written any curve record (the tool writes records at the end); it was rerun from scratch with the same command and seed, and the interruption is recorded in `env.txt`. No other arm was interrupted.

Each model is censused over **all** its structures, so every per-curve minimum is exact. A, B, D (and R1) cover every K-isomorphism class and give isogeny-class minima; C and E are fixed-seed samples of curves (with their 2-isogeny neighbours), so their fractions are sample counts and class minima are not computed.

n = 3: the mod-4 pattern at q = 5 … 17

Curves (isomorphism classes) by minimal genus. "self" = the curve, "≤1 step" = the curve and its K-rational 2-isogeny neighbours, "class" = all curves with the same `#E(K)`.

field (coverage)	#E(K)	curves	self	≤1 step	class
q = 5 (all, = merged)	odd	84	13: 84	13: 84	13: 84
	2 mod 4	64	5: 60, 9: 4	5: 60, 9: 4	5: 62, **9: 2**
	0 mod 4	104	3: 62, 5: 42	3: 104	3: 104
q = 7 (all, = merged)	odd	232	13: 232	13: 232	13: 232
	2 mod 4	172	5: 168, 9: 4	5: 168, 9: 4	5: 172
	0 mod 4	286	3: 157, 5: 126, 9: 3	3: 286	3: 286
q = 11 (all, panel A)	odd	888	13: 888	13: 888	13: 888
	2 mod 4	666	5: 660, 9: 6	5: 660, 9: 6	5: 666
	0 mod 4	1108	3: 551, 5: 554, 9: 3	3: 1100, 5: 8	3: 1108
q = 13 (all, panel B)	odd	1468	13: 1468	13: 1468	13: 1468
	2 mod 4	1100	5: 1092, 9: 8	5: 1092, 9: 8	5: 1098, **9: 2**
	0 mod 4	1832	3: 888, 5: 938, 9: 6	3: 1832	3: 1832
q = 17 (sample of 200, panel C)	odd	67	13: 67	13: 67	not computed
	2 mod 4	55	5: 55	5: 55	not computed
	0 mod 4	78	3: 32, 5: 46	3: 77, 5: 1	not computed
q = 3 (all, pilot, char 3)	odd	20	13: 20	13: 20	13: 20
	2 mod 4	14	5: 12, 9: 2	5: 12, 9: 2	5: 12, 9: 2
	0 mod 4	22	3: 14, 5: 6, 7: 1, 9: 1	3: 20, 7: 2	3: 20, **7: 2**



n = 3: minimal genus by #E(K) mod 4 (q = 5, 7 reproduce PR #1759)

Prediction P3 (pre-registered) is falsified at q = 13, in the 2 mod 4 row only. Odd ⟶ 13 and 0 mod 4 ⟶ 3 hold on every curve at q = 11 and q = 13 (888 + 1468 odd, 1108 + 1832 0 mod 4 curves); 2 mod 4 ⟶ 5 holds on all 666 curves at q = 11 but **2 of 1100 curves at q = 13 have class minimum 9**. The derived statements held everywhere: no odd curve below 13 (every odd structure is 13 or 17), no 2 mod 4 curve below 5.

Observation (post hoc, not predicted; bounded census): in every exhaustive field with p > 3 (q = 5, 7, 11, 13), every curve that misses the mod-4 class pattern lies in the trace-zero isogeny class #E(K) = Q + 1 (Q = q^3): 126 at q = 5 (both merged exceptions) and 2198 at q = 13 (both new exceptions), each with j ∈ F_q and a τ-stable structure (m1 = 1). That class is 2 mod 4 exactly when q ≡ 1 mod 4; for q ≡ 3 mod 4 (q = 7, 11, Q + 1 = 344, 1332) it is 0 mod 4 and does reach genus 3. Restated with this class removed, the pattern 13 / 5 / 3 holds on every curve of the four fields. The q = 3 pilot (characteristic 3, outside the merged scope) has further exceptions: the trace-zero class 28 (0 mod 4, minimum 7) and two 2 mod 4 classes (18, 38) with minimum 9. Why the trace-zero class misses is not derived here.

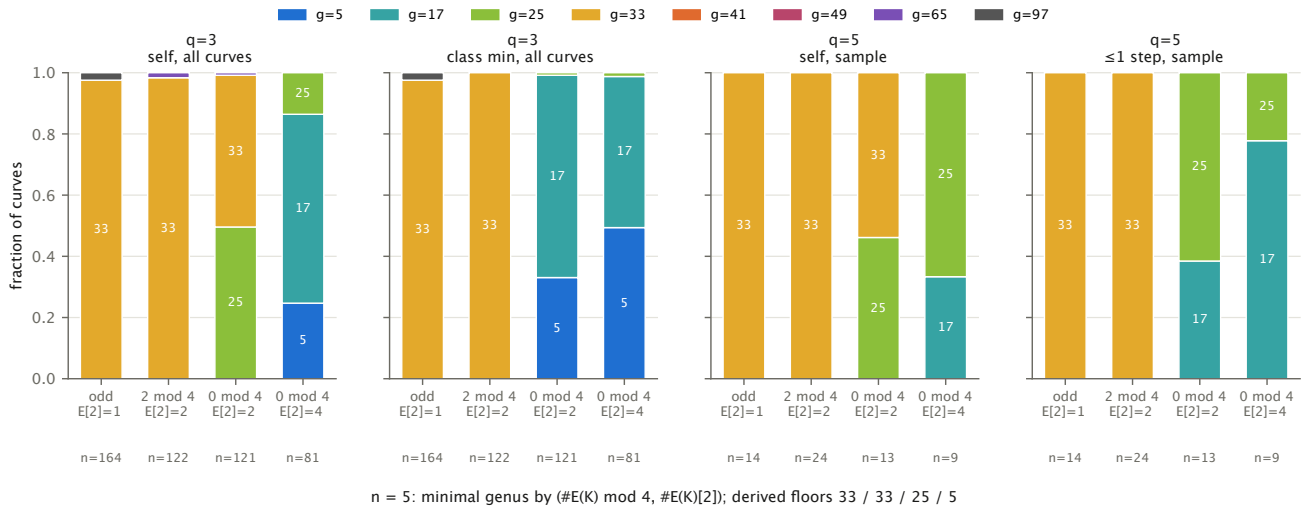
Within one 2-isogeny step, genus 3 is reached by every 0 mod 4 curve at q = 13 and by 1100 of 1108 at q = 11; the other 8 need a longer path (their class minimum is 3). At q = 17, 77 of 78 sampled 0 mod 4 curves reach 3 within one step.

n = 5: what governs the minimal genus

Curves by minimal genus, by 2-torsion type. Derived floor in brackets (Prop. O5, Cor. T5, Lemma E).

field (coverage)	#E(K), E(K)[2] [floor]	curves	self	≤1 step	class
q = 3 (all, panel D)	odd, 1 [33]	164	33: 160, 97: 4	33: 160, 97: 4	33: 160, 97: 4
	2 mod 4, 2 [33]	122	33: 120, 65: 2	33: 120, 65: 2	33: 122
	0 mod 4, 2 [25]	121	25: 60, 33: 60, 65: 1	5: 20, 17: 90, 25: 11	5: 40, 17: 80, 25: 1
	0 mod 4, 4 [5]	81	**5: 20**, 17: 50, 25: 11	5: 40, 17: 40, 25: 1	5: 40, 17: 40, 25: 1
q = 5 (sample of 60, panel E)	odd, 1 [33]	14	33: 14	33: 14	not computed
	2 mod 4, 2 [33]	24	33: 24	33: 24	not computed
	0 mod 4, 2 [25]	13	25: 6, 33: 7	17: 5, 25: 8	not computed
	0 mod 4, 4 [5]	9	17: 3, 25: 6	17: 7, 25: 2	not computed

Finer 2-adic data at q = 3 (class minimum by v_2(#E(K))): v_2 = 0 ⟶ 33 (160) or 97 (4); v_2 = 1 ⟶ 33 (122); v_2 = 2 ⟶ 17 (80), 25 (2); v_2 = 3 ⟶ 5 (40), 17 (20); v_2 = 4 ⟶ 5 (40); v_2 = 5 ⟶ 17 (20). On the curve itself genus 5 occurs only with #E(K)[4] = 8 (E(K)[2^∞] ≅ Z/2 × Z/4; 20 of 60 such curves, 0 of 21 with #E(K)[4] = 4).



n = 5: minimal genus by (#E(K) mod 4, #E(K)[2]); derived floors 33 / 33 / 25 / 5

What governs the n = 5 minimum is the 2-torsion Galois type, sharpened by #E(K) mod 4: a derived floor for each type, attained in the data.

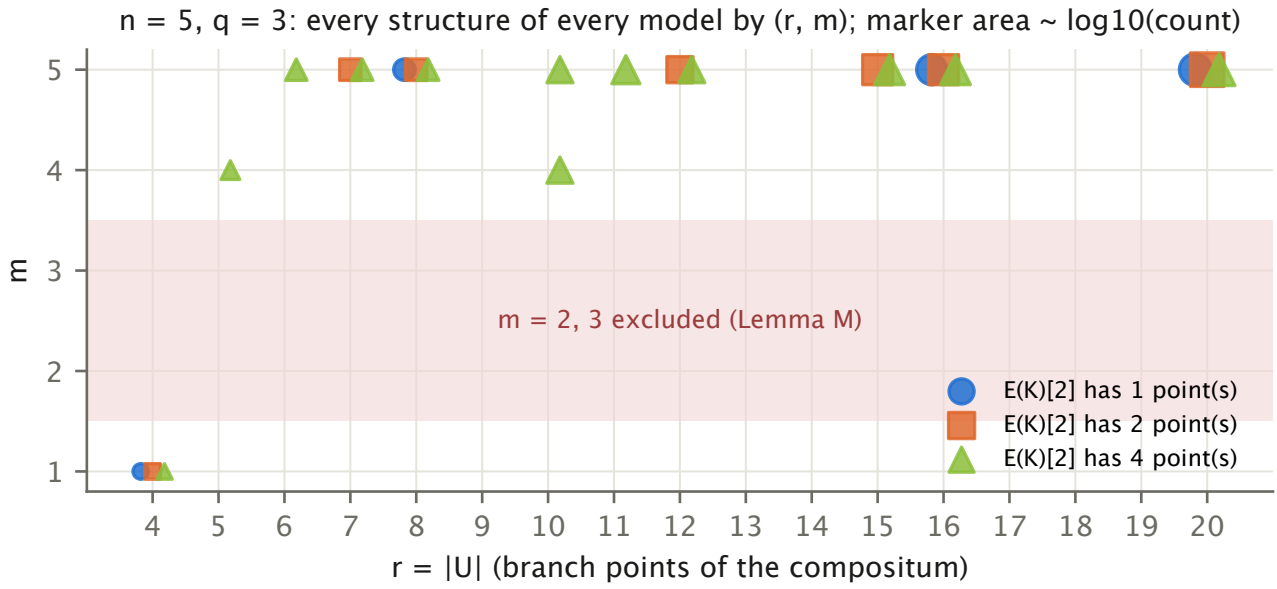
- *Odd order* (E(K)[2] = 0): minimum 33 for every curve with j(E) ∉ F_q (160 at q = 3, 14 of 14 sampled at q = 5), attained by exactly one structure, as Proposition O5 says; 97 for the 4 odd curves at q = 3 with j ∈ F_q (descended, m1 = 1). Derived and confirmed; no isogeny changes it.
- *2 mod 4* (E(K)[2] = Z/2, T ∉ 2E(K)): floor 33 (Cor. T5), and the class minimum is 33 on all 122 curves at q = 3; 24 of 24 sampled at q = 5 have 33 on the curve itself.
- *0 mod 4 with E(K)[2] = Z/2*: floor 25 on the curve itself (Cor. T5), reached by 60 of 121 at q = 3 and 6 of 13 sampled at q = 5. One 2-isogeny step moves them to split curves: class minimum 5 for 40 and 17 for 80 of 121 at q = 3.

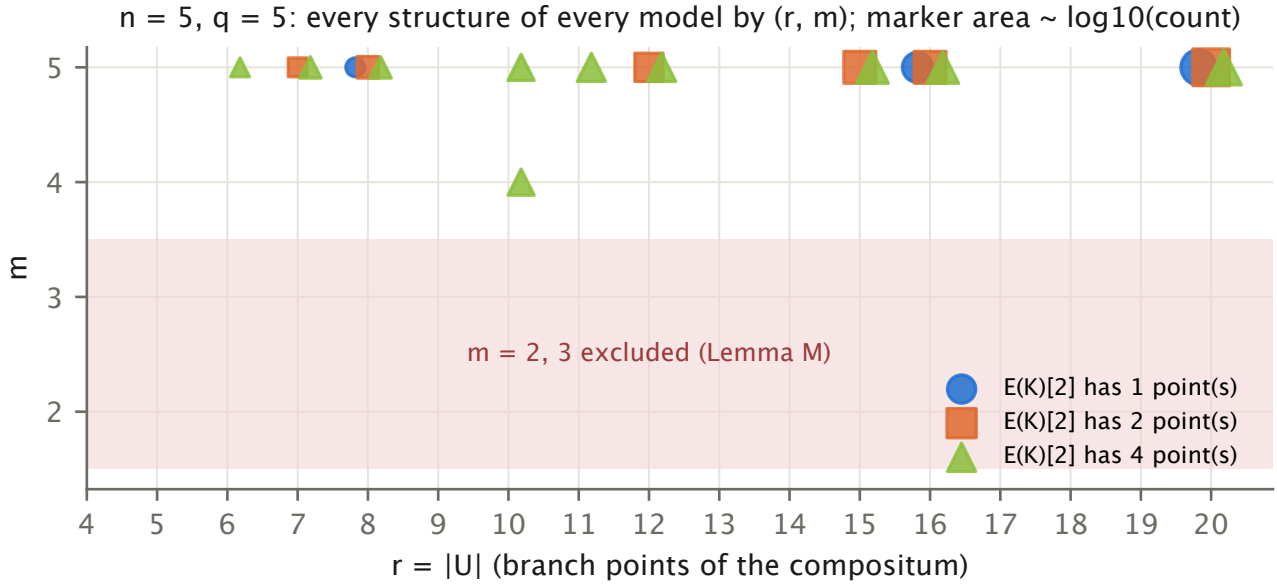
- *Split 2-torsion*: only these can reach genus 5 (Lemma E), via `B ~ O(x) \ {σ^j x}` (four of the five conjugates of a point `x ∈ K \ F_q`). At `q = 3`, 20 of 81 split curves reach 5 on themselves and 40 of 81 in their class. **At `q = 5` none of the 9 sampled split curves, and none of the 31 split models censused including neighbours, reaches 5** (minimum 17). Expected if the genus-5 configurations are rare for larger `q`: they form a family of about $5(q^5 - q)/(q^3 - q) \approx 5q^2$ structures, against about $Q = q^5$ curves (a heuristic count, not a derivation). With 9 sampled split curves this is a small-sample statement, not an estimate of the rate.

Hypothesis H (registered after panel D, before panel E's output): at `q = 3`, `E(K)[2] = Z/2` curves reach 25 on themselves only with $v_2 \geq 3$; the prediction was that this restriction disappears at `q ≡ 1 mod 4`. At `q = 5`, 5 of the 8 sampled `E(K)[2] = Z/2`, $v_2 = 2$ curves have genus 25 ($8 \geq 5$, so the test was powered): **H held**, which fits the exponent explanation in the amendment. It is still unproved.

So the answer to "does `n = 5` follow `#E(K) mod 4`?" is no, not in the `n = 3` sense. Odd and 2 mod 4 sit at the same derived value (33) and never go lower. The 0 mod 4 class splits by the 2-torsion field (25 vs 5 on the curve itself) and, over the isogeny class, reaches 5 or stops at 17 (`q = 3`, exhaustive) for reasons the census does not explain: within 0 mod 4 at `q = 3` the class minimum follows v_2 only loosely (table above). The trace-zero class `#E(K) = Q + 1 = 244` at `q = 3` stops at 25.

Structures by `(r, m)` over all models (`mr` records): `m ∈ {1, 4, 5}` only, `m = 4` only for split 2-torsion and `r ∈ {5, 10}`, as Lemma M and Lemma E require.





At `q = 5` the configuration `(m, r) = (4, 5)` (genus 5) does not occur in any of the 124 censused models, while `(4, 10)` (genus 25) does.

Derivations (full statements and proofs in PROTOCOL.md, "Derived statements")

All were written into the protocol before the panels ran; the tool checks each on every curve (`violations`, `derived_checks`): **0 violations in every arm.**

statement	content	status
Lemma M	span of {σ^i 1_{B'}} is a cyclic F_2[x]/(x^n−1)-module: m ∈ {1,2,3} (n = 3), **m ∈ {1,4,5} (n = 5)**, m = n−1 iff every point of U lies in an even number of the σ^iB'	proved (paper); Lean: the single-orbit case Rot5.span_sizes
Lemma D0	n prime, m ≥ 2 ⟹ g ≥ n	proved (paper)
Lemma E	m = n−1 forces all four branch points K-rational; n = 5: g ∈ {5, 25}; n = 3: (m,r) = (2,6)	proved (paper)
Corollary	n = 5 genera are 5, 25 or 8(r−4)+1 ≥ 17: **nothing in 6–16; genus 5 only if E(K)[2] = (Z/2)²**	proved (paper); Lean: arithmetic Spectrum.n5_gap
Prop. O5	odd #E(K), n = 5: genera in {33, 97, 129}; **exactly one structure** has θ-orbit 3, and it is m = 1 iff B descends; for j ∉ {0,1728}: **minimum 33 iff j(E) ∉ F_q, else ≥ 97**	proved (paper); Lean: orbit step Orbit.period_three_or_fifteen
Prop. T	type (1,1,2): χ = [cr(a₁,a₂;β,βʃ)] ∈ U₁/U₁² is trivial iff 4 | #E(K); the configuration {β,β^q} ∪ {x, σ^kx} forces χ = 1	proved (paper); Lean: exponent parity Parity.even_one_sub_neg_pow
Cor. T3	**n = 3: #E(K) ≡ 2 mod 4 ⟹ every structure has g ≥ 5** (proof of the merged census's observation)	proved (paper)
Cor. T5	n = 5: E(K)[2] = Z/2 ⟹ g ≥ 25; #E(K) ≡ 2 mod 4 ⟹ g ≥ 33	proved (paper)
Lemma H	m ≥ 3: hyperelliptic iff an involution of the cover group has genus-0 quotient (exact, not just sufficient); m ≥ 4 never hyperelliptic	proved (paper)

So for n = 3 the merged census's "2 mod 4 never reaches 3" and "odd ≥ 13" are now both theorems (for every q), and its observation that genus 3 needs 4 | #E(K) follows: every genus-3 configuration is split or carries the trivial cross-ratio class of Proposition T. For n = 5 the floors by 2-torsion type are 33 (odd), 33 (2 mod 4), 25 (E(K)[2] = Z/2, 4 | #E(K)) and 5 (E(K)[2] = (Z/2)²); the census shows which are attained.

Lean (CoverCensusN5.lean, toolchain v4.34.1, no Mathlib, `lake build` log in lean/build.log):

report statement	Lean declaration	status / axioms
Prop. O5 orbit step: τ¹⁵θ = θ, τ⁵θ ≠ θ ⟹ returns before 15 are multiples of 3, τθ ≠ θ	Orbit.period_three_or_fifteen (via Orbit.ret_gcd)	proved; propext, Quot.sound
merged OrbitNine re-derived	Orbit.no_short_return_nine	proved; propext, Quot.sound
Lemma M for one free 5-orbit (span sizes 1, 2, 16, 32; 16 iff rotations sum to 0)	Rot5.span_sizes (decide, kernel-checked)	proved; propext
Prop. T exponent 1 − (−q)^k even	Parity.even_one_sub_neg_pow	proved; propext, Quot.sound
n = 5 genus spectrum arithmetic	Spectrum.n5_gap	proved; no axioms
Lang reduction, Lemma M for arbitrary σ-sets, Lemma E, simple transitivity in O5, the cross-ratio invariant and 2-descent in T, genus formula, Lemma H	none	**not formalized** (needs field, module and curve theory absent without Mathlib)

No `sorry`, no custom axioms, no `native_decide`. No identity certificate: no polynomial identity is claimed.

What this does and does not say

- **Measured, bounded:** the n = 3 pattern at q = 11, 13 (every curve) and q = 17 (sample of 200); n = 5 at q = 3 (every curve) and q = 5 (sample of 60 plus 64 neighbours). Every model is censused over all of its structures. A bounded census is not a theorem: the trace-zero observation and the n = 5 attainment rates are observations.
- **Proved (paper proofs in PROTOCOL.md; Lean for the orbit, single-orbit module, parity and spectrum steps):** the floors and the spectrum listed under Derivations. These hold for every odd q (char ≠ 2) and do not depend on the census. They rest on the merged census's construction (branch-set genus formula), which this study does not re-derive.
- **Cost context (published asymptotics, not measured):** genus 3 is below the rho exponent for n = 3 (merged PROTOCOL.md references). For n = 5, rho on E(K) costs about q^{5/2}; index calculus on a genus-g Jacobian over F_q costs about q^{2−2/g} (Diem 2011), which is below q^{5/2} for every g. So even genus 33 is "sub-rho" in the exponent, but the constants grow very quickly with g. This study prices nothing and makes no attack claim.
- **Not established:** that the covers descend to F_q with a transfer injective on the large prime subgroup (cited for odd n, not verified); n = 4 (descent not decidable from branch data, left out, PROTOCOL.md); degree-2 maps other than x; why the trace-zero class misses the n = 3 pattern; any reason for the q = 3 v₂ ≥ 3 restriction beyond the heuristic exponent argument; how often genus 5 occurs at n = 5 for q ≥ 5; hyperellipticity of n = 3, m = 2 covers.

- **Next discriminating step:** an exhaustive, or much larger, split-curve census at `n = 5`, `q = 5`, targeted at the genus-5 family `B ~ O(x) \ {σ^j x}`. Enumerating `x` up to `PGL2(F_q)` (about q^2 orbits) gives every genus-5 curve directly, without the q^{12} structure scan. That would say whether genus 5 at `n = 5` thins out like `q^{-3}`.

Graphs checked and left unchanged

No canonical graph changes: no ECDLP rung, no timing claim, no cost ratio, so no dashboard, leaderboard or `benchmarks/` entry (wall times in `env.txt` are bookkeeping on a shared host, not performance results). The merged census's figures are unchanged; this study's figures are new.

Reproduce

```
sh run.sh 20261014T0050Z <bin-dir>          # with STOP_AFTER_REPRO=1: pilot, R1, R2, panel D, R3
sh compare.sh 20261014T0050Z <scratch-dir>  # R1-R3 (jq + cmp)
sh run_panels.sh 20261014T0050Z <bin-dir>   # panels A, B, C, E
python3 reports/plot.py results/run-20261014T0050Z reports/run-20261014T0050Z   # renders recorded values
only
cd lean && lake build
typst compile REPORT.typ
```